INCH-POUND

MIL-C-43613D
8 December 1993
SUPERSEDING
MIL-C-43613C
14 April 1986

MILITARY SPECIFICATION

CAN, WATER, MILITARY: PLASTIC, 5-GALLON CAPACITY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers a 5-gallon capacity, rectangular, plastic, military water can with carrying handle.

1.2 Classification. The plastic 5-gallon water containers shall be of the following types (see 6.2):

Type 1 - Tan
Type 2 - Green

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be used in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 7240

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SPECIFICATIONS

FEDERAL

L-P-390	-	Plastic, Molding and Extrusion Material, Polyethylene and Copolymers (Low, Medium, and High Density)
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes

STANDARDS

FEDERAL

FED-STD-595	-	Colors

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

2-9-248	-	Can, Water, Military, Plastic, 5-Gallon Capacity; Assembly
2-9-249	-	Can, Water, Military, Plastic, 5-Gallon Capacity; Cap Assembly
2-9-285	-	Adapter, Assembly and Details
10-1-487	-	Can, Water, Plastic, 5-Gallon Capacity; Packing for Level B Shipment

(Copies of drawings are available from the U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-UXT, Natick, MA 01760-5017.)

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U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES (HHS), U.S. FOOD AND DRUG ADMINISTRATION (FDA)

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1-199)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to the first article inspection (see 6.3) in accordance with 4.3.

3.2 Federal Food and Drug Act certification. Prior to the approval of the first article, if one is submitted, or prior to approval of the first shipment, the contractor shall submit satisfactory evidence to the contracting officer or his authorized representative, that the plastic for the water can and the antioxidants, light stabilizers, or other substances incorporated in the plastic proposed to be supplied under this specification meets the requirements of 21 CFR - Food and Drugs: Indirect Food Additives: Polymers, Subpart B, Part 177, Chapter I, Title 21, Code for Federal Requirements for Contact with Food. Acceptable evidence of meeting the requirements of this document shall be a certified test report from a recognized independent testing laboratory, acceptable to the Department of the Army Surgeon General. This certification shall be required for all acquisitions without exception.

3.3 Materials.

3.3.1 Polyethylene, high density. The plastic body shall consist of a high density polyethylene copolymer resin conforming to type I, class B of L-P-390 plus a light stabilizer, and requirements of 3.2 except that the basic resin prior to adding the light stabilizer shall have a density of not less than 0.954 and not more than 0.956, and a melt index of not more than 0.3.

3.3.2 Polyethylene, low density. The complete plastic cap and strap shall be low density virgin polyethylene conforming to type I, class L of L-P-390 plus a light stabilizer and requirements of 3.2.

3.4 Design and construction. The design and construction of the water container shall be as shown on the drawings specified in 2.1.2 and as specified herein. Silicone-type mold release agents shall not be used with

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polyethylene. Mold release agents which are not active stress-cracking agents for polyethylene shall be used. A certificate of compliance shall be furnished stating that the mold release agents comply with this requirement (see 4.5). A certificate of compliance shall be furnished stating that the material with the light stabilizers used on the complete cap and body meet the test requirements of 4.6.4 (see 6.5.2). The male and female threads shall be fully formed and free from flash or thread misalignment at the parting line. The finished water can shall not leak when tested as specified in 4.6.1.

3.4.1 Body. The body shall be fabricated from the high density polyethylene specified in 3.3.1, and shall consist of a bottom, four sides, and top with handle and a body neck and shall be formed by blow molding to give a one piece construction. The body walls shall be a minimum of 0.100 inch thick, except that the thickness of the four top outside curved surfaces of the handle may be 0.075 inch minimum; however, the 0.075 inch minimum thickness shall not extend over an area greater than 0.25 inch in diameter. The body neck shall have a thickness of not less than 0.151 inch in all areas below the area finished to form the liquid seal. The body walls shall not crack or break when tested as specified in 4.6.2. The upper surface of the body neck that comes in contact with the tapered sealing surface on the inside of the cap to form a seal shall be flat and smooth and may be out of round, in a plane parallel to the top surface of the body neck, provided that the fit or water tightness is not impaired. The interior surface of the body shall be smooth. The interior rounded surface area where the body and the neck join shall have a uniform texture having no rough areas, surface abnormalities, surface deformations, or surface irregularities exceeding those on the standard sample in magnitude and frequency (see 6.4), when compared with the standard sample.

3.4.2 Cap. The cap assembly components, except the retaining pin, shall be injection molded and fabricated of the material specified in 3.3.2. The retaining pin shall be fabricated from the material specified in 3.3.1 and may be sealed to the breather stud cap using a heating method that does not leave burn marks. The water can cap shall lock in the closed or tightened position by means of a tab on the water can cap assembly which is engaged to the raised tab located at the base of the body pouring neck on the body. The breather stud cap shall be secured to the breather stud in the closed or tightened position by means of a retaining pin inserted through both the breather stud and breather stud cap. The breather stud shall be free and clear of all mold flashes. The caps or cap areas shall not crack or break when the water can is tested in accordance with 4.6.2. The plastic caps shall not crack when tested in accordance with 4.6.3. The upper surface of the cap pouring spout neck, that comes in contact with the tapered sealing surface on the inside of the pouring spout cap to form a seal, shall be flat and smooth and may have an out of roundness in a plane parallel to the top surface of the pouring spout provided that the fit or water tightness is not impaired.

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3.5 Color. The through color of the material for the body, caps, and breather after molding shall approximate color No. 30279 (sand, lusterless) for Type I water cans and color No. 34094 (green, lusterless) for Type II water cans of FED-STD-595. The complete cap and breather shall match the shade of the body.

3.6 Finish. The exterior surfaces of the water can and the cap assembly shall have a smooth and lusterless finish throughout. All surfaces shall be free of dirt, dust, and foreign matter inclusion. The finish shall be produced from a die that has cavity surfaces finished in vapor blast, water hone, or similar satin finish, but not by the application of any protective coating, lacquer, or other materials. The inside of the cap assembly, including threaded portion, shall possess a uniformly high luster throughout as imparted by a polished die.

3.7 Marking. The water can shall be permanently marked "Property of the U.S. Government" and "WATER" by an embossed molded-in process. The water can shall be marked by an embossed molded-in process with the year of manufacture and the contractor's name or trademark. The container cap shall be marked with raised characters by an embossed molded-in process with the manufacturer's name or trademark.

3.8 Workmanship. The water can shall be clean, void of plastic shavings, and free from flash, bubbles, cuts, tears, holes, burns, breaks, cracks, pinholes, pins, warpage, blisters, scratches, or any weld marks or sink marks, except that mold parting lines are acceptable. The water can shall not show any signs of discoloration, surface deterioration, roughness, orange peel, or grease.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice

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to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.2.1 and the specified dimensions, and tested in accordance with 4.6.1, 4.6.2, 4.6.3, and 4.6.4. Failure to pass any test shall be cause for rejection of the first article. The presence of any defect or failure to pass any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 In-process inspection.

4.4.2.1 In-process examination. Examination shall be made of the fabrication processes in table I to establish conformance with specified requirements. Whenever noncompliance is noted, correction shall be made to the applicable process and all items processed.

TABLE I. In-process examination

Requirement	Reference paragraph
Blow molding of can body	3.4 and 3.4.1
Injection molding	3.4.2
Cavity surfaces of finished die	3.6

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4.4.3 End item inspection. The inspection lot shall be all water cans offered for inspection at one time. The sample unit shall be one complete water can.

4.4.3.1 End item visual examination. The end items shall be examined for the defects listed in table II. The lot size shall be expressed in units of cans. The sample unit shall be one can. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

TABLE II. End item visual defects

Examine	Defect	Classification Major	Minor
Finish	Color not as specified	101	
	Exterior surface of body, cap, or strap not smooth and not lusterless throughout (except any die parting lines)	102	
	Inside surface of cap, including threaded portion, not smooth		201
	Inside surface of cap, including threaded portion, not of uniform high luster		202
Construction and workmanship (applicable to all assemblies and components)	Component missing	103	
	Any bubbles, cuts, tears, holes, flash, burns, breaks, cracks, pinholes, pits, warpage, blisters, scratches, welds or sink marks, or not clean or smooth	104	
	Any discoloration, surface deterioration or roughness, foreign inclusion, or orange peel	105	
Body	Sealing surface of lip is not flat, smooth, and free from sink marks or depressions	106	
	Interior of neck opening not smooth	107	
	Interior surfaces of the body not smooth	108	
Cap and strap (large cap or spout cap as applicable)	Sealing surface in cap is irregular, not smooth, or contains high or low points	109	
	Strap locking tab does not engage with tab on body in closed position	110	
	Cap does not screw on body neck or pouring spout	111	

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TABLE II. End item visual defects (cont'd)

Examine	Defect	Classification Major	Classification Minor
Cap and strap (large cap or spout cap as applicable) (cont'd)	Cap bands when screwed on body neck or pouring spout		203
	Strap does not engage in retention groove		204
	Breather air vent holes not clear	112	
Threads	Threads in caps, body neck, pouring spout, or breather stripped	113	
	Threads not fully formed, not free from flash, or misaligned at parting line	114	
Marking	Missing, incomplete, not legible, not specified type or size, or not in proper location		205

4.4.3.2 End item dimensional examination. The end items shall be examined for conformance to the dimensions annotated with an asterisk on Drawing 2-9-248. Only those dimensions that can be evaluated without damaging or disassembling the end items shall be examined. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of cans. The sample unit shall be one can. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.3.3 End item testing. The end items shall be tested as specified in 4.6.1, 4.6.2, and 4.6.3 to detect the defects specified in table II. The lot size shall be expressed in units of cans. The sample unit shall be can. The inspection level shall be S-3, and the AQL, expressed in terms of defects per hundred units, shall be 2.5. Any test failure shall be cause for rejection of the lot.

4.4.4 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

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Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Content	Number per container is more or less than required

4.4.5 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Load not bonded with required straps specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.5 Certification examination. Certificates of compliance, certified test reports, approval labels or listing marks for codes and standards, as applicable, that are submitted as proof of compliance with the document requirements shall be examined and validated for mold release agents and light stabilizers (see 3.2 and 3.4).

4.6 Test methods.

4.6.1 Leakage test. The water can shall be filled with water at 73°F ± 3°F. The cap shall be screwed down and tightened, and the water can shall be allowed to remain on its bottom for 10 minutes and examined for compliance with 3.4. The exterior of the water can and cap shall be free of any moisture. The water can shall be rotated 180° so as to be resting on

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its handle, with the cap facing down for not less than one-half hour and examined for any leakage or presence of moisture. To determine compliance with 3.4, any evidence of leaking or presence of moisture shall constitute failure.

4.6.2 Drop test. The water can shall be filled with 5 gallons of water at 73°F ± 3°F. The filled water can shall be conditioned for 24 hours at 0°F. The conditioned water can shall then be dropped from a height of 6 feet onto a solid concrete horizontal surface in such a manner that the base of the container strikes squarely against the surface to determine compliance with soundness requirements in 3.4. After the above drop test has been accomplished, the conditioned water can shall be allowed to regain room temperature and then examined for leakage as specified in 4.6.1. Any evidence of leakage shall constitute failure of this test.

4.6.3 Environmental stress cracking tests for caps. The neck threads on the body and the cap threads, including the area between the base of threads and sealing ring, shall be thoroughly coated with full strength Igepal CO-630 (see 6.5.1). The cap shall be screwed down on the body and tightened to a torque of not less than 15 foot pounds by the use of a torque wrench utilizing the adapter shown on Drawing 2-9-285. The container shall then be placed in an upright position in a circulating air area at a temperature of 140°F ± 2°F. After 30 hours exposure, the container shall be removed from the oven and the cap examined for cracks. If the cap shows no cracks, recheck the torque. If it is less than 15 foot pounds, retighten the cap to a minimum of 15 foot pounds and re-examine the cap immediately for cracks. The pouring cap shall be screwed to the pouring spout and the breather shall be open during the period of the test. Any visible evidence of cracks or cracking shall constitute failure of this test.

4.6.4 Light stabilizer tests for materials used in the body and complete cap. Plastic samples (compression molded plaques) shall be exposed to a 6,000 watt Xenon Arc Weatherometer for 130, 24-hour days, straight running time. There shall be four one-half hour moisture cycles a day. At the completion of the tests, the samples shall show no signs of deterioration or degradation.

5. PACKAGING

5.1 Packing. Packing shall be level A or B, as specified (see 6.2).

5.1.1 Level A packing. Seven water cans shall be packed as shown on Drawing 10-1-487.

5.1.2 Level B packing. Seven water cans shall be packed as shown on Drawing 10-1-487.

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5.2 Palletization. Water cans, packed as specified, shall be prepared in a unitized pallet load in accordance with load type VIII of MIL-STD-147, except that the overall height of the load shall not exceed 43 inches. The palletized load shall be provided with a fiberboard cap and shall be bonded with primary and secondary straps and horizontal straps centered in each course. A horizontal separator is not required. The fiberboard cap shall be fabricated with V2s of PPP-F-320.

5.3 Marking. In addition to any special marking required by the contract or purchase order, palletized unit loads shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The water cans specified herein are used for supplying potable water to personnel of the Armed Services in zones I through VI, but not in zone VII (Arctic).

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type of water can desired (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.3, and 6.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of Federal Acquisition Regulation (FAR) 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Standard sample. To determine the degree of smoothness required on the interior surface where the neck and body join (see 3.4), a standard sample showing an acceptable smoothness may be obtained from the Director, Technical Operations, Defense Logistics Agency, Defense General Supply Center, Richmond, VA 23297.

6.5 Commercial sources.

6.5.1 GAF Corporation. Igepal CO-630, required in 4.6.3, may be obtained from GAF Corporation (FSCM 24299), 1361 Alps Road, Wayne, NJ 07470.

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6.5.2 Ciba Geigy Corporation. Tinuvin 622, a product of Ciba Geigy Corporation, Plastics and Additive Division, 7 Skyline Drive, Hawthorne, NY 10532, has been found to meet the requirements of 3.4.

6.6 Subject term (keyword) listing.

Container, polyethylene
Container, water
Jug, water
Polyethylene container

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians

Army - GL
Navy - SA
Air Force - 99

Preparing activity:

Army - GL

Project 7240-0260

Review Activities

Army - MD
Navy - CG
Air Force - 84
DLA - GS

User Activities

Navy - YD, MC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revisi letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-43613D	8 December 1993

3. DOCUMENT TITLE
Can, Water, Military· Plastic, 5-Gallon Capacity

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

[redacted]	[redacted]	[redacted]

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE *(Include Area Code)*	
U.S. Army Natick RD&E Center	(1) Commercial 508-651-5175	(2) AUTOVON /DSN 256-5175
c. ADDRESS *(Include Zip Code)* Commander, U.S. Army Natick RD&E Center ATTN: SATNC-WEE Natick, MA 01760-5018	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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